

Chapter 2: Data Communication Fundamentals

Unit 2.5

Data Rate Limits

The maximum data rate limit over a medium is decided by the followings:

- Bandwidth of the channel
- Signal Levels
- Channel quality (level of noise)

Channel Capacity

The maximum rate at which data can be transmitted over a given communication path

Two theoretical formulas were developed to calculate the data rate.

1. Nyquist Bit Rate (for noiseless channel)

The Nyquist Theorem applies to noiseless channels and sets a limit based on bandwidth and signal levels.

$$\text{Bit rate (R)} = 2 * \text{Bandwidth} * \log_2 L \quad \text{bits/sec}$$

Where, L = Number of discrete signal levels (e.g., 2 for binary)

2. Shannon Capacity (for noisy channel)

$$\text{Capacity} = \text{Bandwidth} * \log_2(1 + \text{SNR}) \quad \text{bits/sec}$$

Where, SNR = Signal to Noise Ratio

- ❖ A noiseless communication channel has a bandwidth of 4000 Hz and transmits a signal using 4 distinct signal levels. What is the maximum achievable bit rate for this channel?

Solution:

$$\begin{aligned} \text{Given, Bandwidth (B)} &= 4000\text{Hz} \\ \text{Signal Levels (L)} &= 4 \\ \therefore \text{Bit Rate (R)} &= 2 * B * \log_2(L) \\ &= 2 * 4000 * \log_2 4 = 16000 \text{ bits/sec} \end{aligned}$$

- ❖ Calculate the theoretical highest bit rate of a regular telephone line. A telephone line normally has a bandwidth of 3000Hz. The signal-to-noise ratio is usually 3162.

Solution:

$$\begin{aligned} \text{Given, Bandwidth (B)} &= 3000 \text{ Hz} \\ \text{SNR} &= 3162 \\ \text{Capacity} &= B * \log_2(1 + \text{SNR}) = 3162 * \log_2(1 + 3162) = 34881 \text{ bps} \end{aligned}$$

Unit 2.6 : Performance of Network

Network performance refers to the evaluation of a network's service quality as experienced by the end user. It can be assessed through various metrics, depending upon the nature and design of the network.

Parameters for measuring network performance

1. **Bandwidth**
The maximum amount of data that can be transmitted over a network in a given time. The term bandwidth can be interpreted in two distinct contexts, each with its own measurement unit. For digital systems, bandwidth is typically expressed in bits per second (bps) or bytes per second, while for analog systems, it is measured in Hertz (Hz).
2. **Throughput**
Throughput refers to the actual speed at which data is successfully delivered over a network or communication channel during a given time frame. Often called the effective data rate or payload rate, it represents the practical performance of the network under real operating conditions.
3. **Latency (Delay)**
Latency is defined as the total time taken for a complete message to arrive at the destination. Latency is made of four components: propagation time, transmission time, queuing time, and processing delay.
4. **Jitter**
Jitter is the variation in packet arrival time. It affects real-time applications like video calls and online gaming.
5. **Error Rate**
The rate at which errors occur during data transmission. It is measured as bit error rate (BER).